

8. Taylor Polynomials

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Second Partial Derivatives

Consider $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by $f(x, y) = x^2 \cos y$. Then

$$\frac{\partial f}{\partial x} = 2x \cos y \quad \frac{\partial f}{\partial y} = -x^2 \sin y$$

we can differentiate again :

$$\frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) = 2 \cos y \quad \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) = -2x \sin y$$

$$\frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) = -2x \sin y \quad \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right) = -x^2 \cos y$$

These are the second partial derivative
of f .

Notation

There are two main ways to notate second partial derivatives:

- $\frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) = (f_x)_x$ is denoted by $\frac{\partial^2 f}{\partial x^2}$ or f_{xx} ;
- $\frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) = (f_y)_x$ is denoted by $\frac{\partial^2 f}{\partial x \partial y}$ or f_{yx} ;
- $\frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) = (f_x)_y$ is denoted by $\frac{\partial^2 f}{\partial y \partial x}$ or f_{xy} ;
- $\frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right) = (f_y)_y$ is denoted by $\frac{\partial^2 f}{\partial y^2}$ or f_{yy} .

Note the opposite order of the variables in the different notation for the **mixed derivatives**.

Exercise

$$f_x = \frac{-2xy}{(x^2+y^2)^2} \quad f_y = \frac{x^2-y^2}{(x^2+y^2)^2}$$

Exercise. Find the second partial derivatives of $f(x, y) = \frac{y}{x^2 + y^2}$.

$$\begin{aligned} f_{xx} &= \frac{-2y(x^2+y^2) - 2xy(2y)}{(x^2+y^2)^3} & f_{yx} &= \frac{6xy^2 - 2x^3}{(x^2+y^2)^3} \\ &= \frac{6x^2y - 2y^3}{(x^2+y^2)^3} \end{aligned}$$

$$f_{xy} = \frac{6x^2y - 2x^3}{(x^2+y^2)^3}$$

$$f_{yy} = \frac{6x^2y - 2y^3}{(x^2+y^2)^3}$$

The Mixed Derivatives Theorem

The Mixed Derivatives Theorem

Suppose that $f : \Omega \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}$ and the second partial derivatives of f exist at $\mathbf{x}_0 \in \Omega$. If f , $\frac{\partial f}{\partial x}$, $\frac{\partial f}{\partial y}$, $\frac{\partial^2 f}{\partial x \partial y}$ and $\frac{\partial^2 f}{\partial y \partial x}$ are all continuous on an open set containing $\mathbf{x}_0$, then the partial derivatives at $\mathbf{x}_0$ commute. That is,

$$\frac{\partial^2 f}{\partial x \partial y}(\mathbf{x}_0) = \frac{\partial^2 f}{\partial y \partial x}(\mathbf{x}_0).$$

The above theorem is also known as Schwarz's Theorem, or Clairaut's Theorem .

The assumption of **continuity** is important! The following example is due to Peano:

$$f(x, y) = \begin{cases} \frac{xy(x^2 - y^2)}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0). \end{cases}$$

$= x^3y - xy^3$

$\underline{\underline{=}}$

Peano's Example

First observe that

$$|f(x, y)| = |xy| \left| \frac{x^2 - y^2}{x^2 + y^2} \right| \leq |xy| \frac{x^2 + y^2}{x^2 + y^2}$$

$\rightarrow 0$ as $(x, y) \rightarrow (0, 0)$, so f is cont.

For $(x, y) \neq (0, 0)$, then

$$\begin{aligned} f_x(x, y) &= \frac{(3x^2y - y^3)(x^2 + y^2) - (x^3y - xy^3)(2x)}{(x^2 + y^2)^2} \\ &= \frac{y(x^4 + 4x^2y^2 - y^4)}{(x^2 + y^2)^2} \end{aligned}$$

$$|f_x(x, y)| = \frac{|y| |x^4 + 4x^2y^2 - y^4|}{(x^2 + y^2)^2} \leq 2|y| \rightarrow 0$$

as $(x, y) \rightarrow (0, 0)$

Peano's Example

so f_x is
cont.

$$f_x(0,0) = \lim_{h \rightarrow 0} \frac{f(h,0) - f(0,0)}{h} = 0$$

Similarly, for $(x,y) \neq (0,0)$,

$$f(x,y) = \frac{x(x^4 - 4x^2y^2 - y^4)}{(x^2 + y^2)^2}$$

$$\text{and } |f_y(x,y)| = |x| \frac{|x^4 - 4x^2y^2 - y^4|}{(x^2 + y^2)^2}$$

Also $f_y(0,0)$

$$= \lim_{h \rightarrow 0} \frac{f(0,h) - f(0,0)}{h} = 0, \text{ as } (x,y) \rightarrow 0$$

So f_y is continuous.

Peano's Example

$$f_{xy} = (f_x)y$$

Now,

$$\begin{aligned} f_{xy}(0,0) &= \lim_{h \rightarrow 0} \frac{f_x(0,h) - f_x(0,0)}{h} \\ &= \lim_{h \rightarrow 0} \frac{\frac{h(0+0-h^4)}{(h^2)^2} - 0}{h} \\ &= -1 \end{aligned}$$

Peano's Example

$$\begin{aligned} f_{yx}(0,0) &= \lim_{h \rightarrow 0} \frac{f_y(h,0) - f_y(0,0)}{h} \\ &= \lim_{h \rightarrow 0} \frac{h(h^4 - 0 - 0)}{(h^2)^2} - 0 \\ &= 1 \end{aligned}$$

So $f_{xy}(0,0) \neq f_{yx}(0,0)$

By the Mixed Derivative Theorem, at least one of f_{xy} , f_{yx} is not cont.

Peano's Example

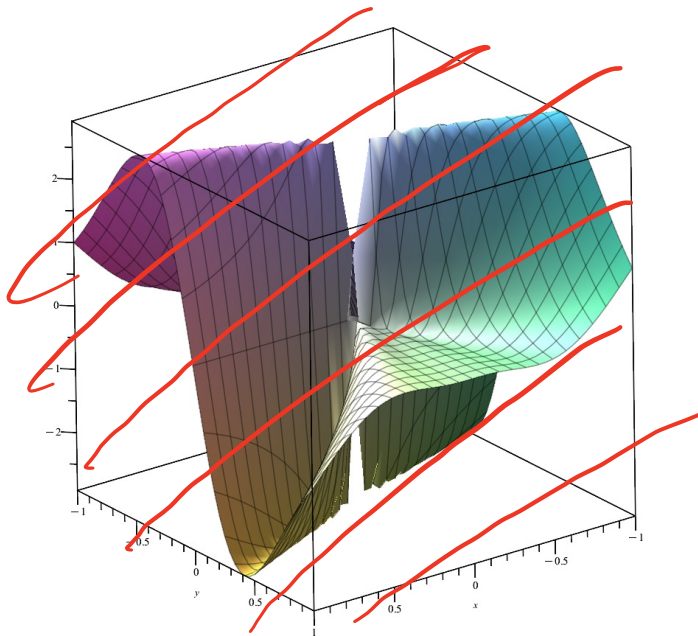
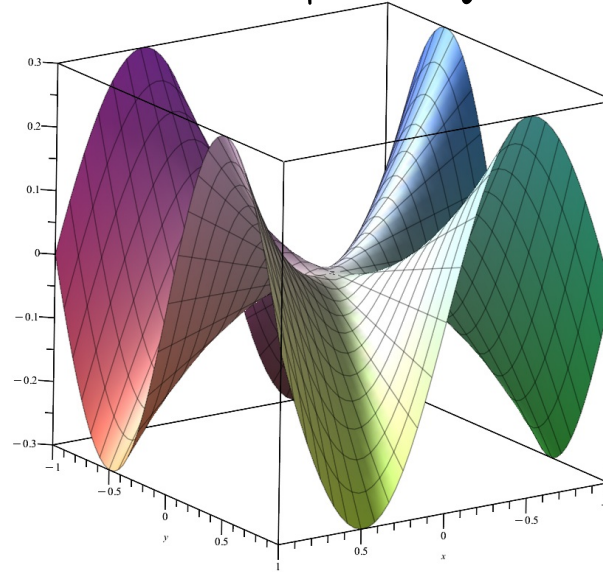
$$f_{xy}(x, y) = \begin{cases} \frac{4x^3y^3 - 12x^5y}{(x^2 + y^2)^2} & \text{if } (x, y) \neq (0, 0) \\ 1 & \text{if } (x, y) = (0, 0) \end{cases}$$

$$f_{yx}(x, y) = \begin{cases} \frac{x^6 + 9x^4y^2 - 9x^2y^4 - y^6}{(x^2 + y^2)^3} & \text{if } (x, y) \neq (0, 0) \\ -1 & \text{if } (x, y) = (0, 0) \end{cases}$$

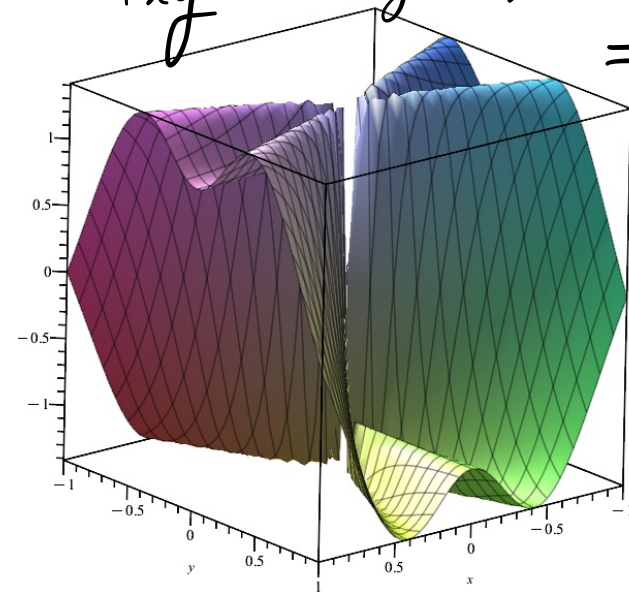
$$f_{yx}\left(\frac{1}{k}, \frac{1}{k}\right) = -1 \quad \nrightarrow \quad 1$$

$$f_{xy}\left(\frac{1}{k}, \frac{1}{k}\right) = 0 \quad \nrightarrow \quad -1$$

$f(x, y)$



$f_{xy} \neq f_{yx}$ for $(x, y) = (0, 0)$



Taylor Polynomials for $f : \mathbb{R} \rightarrow \mathbb{R}$

Recall that for a k times differentiable function $f : \mathbb{R} \rightarrow \mathbb{R}$ and a point $a \in \mathbb{R}$, we can attempt to approximate f near a with the k th order Taylor polynomial of f centred at a :

$$p_{f,k,a}(x) = f(a) + \frac{f'(a)}{1!}(x-a) + \dots + \frac{f^{(k)}(a)}{k!}(x-a)^k$$

where $f^{(j)}(a)$ is the j th derivative of f at a . By design, $p_{f,k,a}^{(j)}(a) = f^{(j)}(a)$ for all $0 \leq j \leq k$.

If $f : \mathbb{R} \rightarrow \mathbb{R}$ is $(k+1)$ times differentiable at $a \in \mathbb{R}$, then one form of Taylor's theorem is that there is $c \in \mathbb{R}$ between a and x such that

$$f(x) = p_{f,k,a}(x) + \frac{f^{(k+1)}(c)}{(k+1)!}(x-a)^{k+1}, \quad c \in (a, x).$$

One could say that

Lagrange form

- $f(a) + f'(a)(x-a)$ is the best affine approximation of f at a ;
- $f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2$ is the best quadratic approximation of f at a ;
- $f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3$ is the best cubic approximation of f at a .

Taylor Polynomials in Two Variables of Degree 2

If all partial derivatives of $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ up to order 2 are continuous, how might we approximate f with a polynomial about the point (a, b) ?

We already have the best affine approximation

$$T_{(a,b)} f(x, y) = f(a, b) + f_x(a, b)(x-a) + f_y(a, b)(y-b)$$

and can try adding terms of degree 2:

$$p(x, y) = T_{(a,b)} f(x, y) + r(x-a)^2 + s(x-a)(y-b) + t(y-b)^2$$

xy has degree 2

We would like to choose r , s and t so that the second partial derivatives of p and f at (a, b) are the same.

$\frac{1}{2!} f''(a) (x-a)^2$

comes from $((x-a) + (y-b))^2 = (x-a)^2 + (y-b)^2 + 2(x-a)(y-b)$

Taylor Polynomials in Two Variables of Degree 2

$$p_{xx} = 2r, \quad p_{xy} = p_{yx} = s \quad p_{yy} = 2t$$

$$r = \frac{1}{2} f_{xx}(a, b), \quad s = f_{xy}(a, b), \quad t = \frac{1}{2} f_{yy}(a, b)$$

The polynomial now is

$$p(x, y) = f(a, b) + f_x(a, b)(x-a) + f_y(a, b)(y-b) + \frac{1}{2} \left[f_{xx}(a, b)(x-a)^2 + 2f_{xy}(a, b)(x-a)(y-b) + f_{yy}(a, b)(y-b)^2 \right]$$

or equivalently

$$f(a, b) + f(x, y) + \frac{1}{2} (x - a) \begin{pmatrix} f_{xx}(a, b) & f_{xy}(a, b) \\ f_{yx}(a, b) & f_{yy}(a, b) \end{pmatrix} (x - a)^T$$

$$\underline{x} = (x, y)$$

$$\underline{a} = (a, b)$$

$\underline{x} - \underline{a}$ thinking as row vector

Taylor Polynomials in n Variables of Degree 2

Similarly, if we have a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ with continuous second order partial derivatives, then we could try to approximate f with by a polynomial about $\underline{a} \in \mathbb{R}^n$:

$$p(\underline{x}) = f(\underline{a}) + \sum_{i=1}^n f_{x_i}(\underline{a}) (x_i - a_i) + \sum_{i,j=1}^n (x_i - a_i) (x_j - a_j)$$

Similarly, we will find that

$$c_{ij} = \begin{cases} \frac{1}{2} f_{x_i x_i}(\underline{a}) & \text{if } i = j \\ f_{x_i x_j}(\underline{a}) & \text{if } i \neq j \end{cases}$$

So the polynomial becomes

$$p(\underline{x}) = f(\underline{a}) + \nabla f(\underline{a}) \cdot (\underline{x} - \underline{a}) + \frac{1}{2} (\underline{x} - \underline{a}) \begin{pmatrix} f_{x_1 x_1}(\underline{a}) & \dots & f_{x_1 x_n}(\underline{a}) \\ \vdots & & \vdots \\ f_{x_n x_1}(\underline{a}) & \dots & f_{x_n x_n}(\underline{a}) \end{pmatrix} (\underline{x} - \underline{a})^T$$

Hessian Matrix

Definition

Suppose that $f : \Omega \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ has all second partial derivatives exist at $\mathbf{a} \in \Omega$. Then the **Hessian matrix of f at $\mathbf{a} \in \Omega$** , denoted by $H_f(\mathbf{a})$, is the $n \times n$ matrix defined by $[H_f(\mathbf{a})]_{ij} = f_{x_i x_j}(\mathbf{a})$. That is,

$$H_f(\mathbf{a}) = \begin{pmatrix} f_{x_1 x_1}(\mathbf{a}) & f_{x_1 x_2}(\mathbf{a}) & \cdots & f_{x_1 x_n}(\mathbf{a}) \\ f_{x_2 x_1}(\mathbf{a}) & f_{x_2 x_2}(\mathbf{a}) & \cdots & f_{x_2 x_n}(\mathbf{a}) \\ \vdots & \vdots & \ddots & \vdots \\ f_{x_n x_1}(\mathbf{a}) & f_{x_n x_2}(\mathbf{a}) & \cdots & f_{x_n x_n}(\mathbf{a}) \end{pmatrix}$$

The Hessian could be conventionally defined via $[H_f(\mathbf{a})]_{ij} = f_{x_j x_i}(\mathbf{a}) = \frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{a})$ instead, but it won't matter when the partial derivatives commute.

Whenever f and its partial derivatives of up to order 2 are continuous, the Mixed Derivatives Theorem implies that $H_f(\mathbf{a})$ is a symmetric matrix.

Hessian Matrix

Example. Recall Peano's example

$$f(x, y) = \begin{cases} \frac{xy(x^2 - y^2)}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0), \\ 0 & \text{if } (x, y) = (0, 0). \end{cases}$$

We calculated earlier that

$$f_x(x, y) = \frac{y(x^4 + 4x^2y^2 - y^4)}{(x^2 + y^2)^2} \quad \text{and} \quad f_y(x, y) = \frac{x(x^4 - 4x^2y^2 - y^4)}{(x^2 + y^2)^2}$$

$$f_{xy}(0, 0) = -1 \quad \text{and} \quad f_{yx} = 1$$

$$f_{xx}(0, 0) = 0 = f_{yy}(0, 0)$$

$$H_f(0, 0) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

Hessian Matrix f has continuous second derivatives

Exercise. Find the Hessian matrix of $f(x, y, z) = e^{xz} + \sin(y^2 z)$ at the point $(-1, 1, \pi)$.

$$f_x = z e^{xz}$$

$$f_y = 2yz \cos(y^2 z)$$

$$f_z = x e^{xz} + y^2 \cos(y^2 z)$$

$$f_{xx} = z^2 e^{xz}$$

$$f_{yy} = 2z \cos(y^2 z) - 4y^2 z^2 \sin(y^2 z)$$

$$f_{zz} = x^2 e^{xz} - y^4 \sin(y^2 z)$$

$$f_{xy} = 0$$

$$f_{yz} = 2y \cos(y^2 z) - 2y^3 z \sin(y^2 z)$$

$$f_{xz} = e^{xz} + xz e^{xz}$$

$$H_f(-1, 1, \pi) = \begin{pmatrix} \pi^2 e^{-\pi} & 0 & (1-\pi) e^{-\pi} \\ 0 & -2\pi & -2 \\ (1-\pi) e^{-\pi} & -2 & e^{-\pi} \end{pmatrix}$$

Differentiability Classes

Definition

If $\Omega \subseteq \mathbb{R}^n$ is open and $f : \Omega \rightarrow \mathbb{R}$, we say that f is (of class) C^k if all partial derivatives of f up to order k exist and are continuous on Ω . The collection of all such functions is denoted by $C^k(\Omega)$.

We also write $C^\infty(\Omega) = \bigcap_{k=1}^{\infty} C^k(\Omega)$ for the collection of **smooth functions on Ω** .

- if $f \in C^1$, then $f(\underline{a}) + \nabla f(\underline{a}) \cdot (\underline{x} - \underline{a})$ is the Taylor polynomial of order 1 about $\underline{a}$.
- if $f \in C^2$, then $f(\underline{a}) + \nabla f(\underline{a}) (\underline{x} - \underline{a}) + \frac{1}{2} (\underline{x} - \underline{a}) H_f(\underline{a}) (\underline{x} - \underline{a})^T$ is the Taylor polynomial of order 2 about $\underline{a}$.

Higher Order Taylor Polynomials

This slide is not examinable!

- If $f \in C^k(\Omega)$, then the polynomial $p_{f,k,\mathbf{a}}$ defined by

$$\begin{aligned} p_{f,k,\mathbf{a}}(\mathbf{x}) = & f(\mathbf{a}) + \nabla f(\mathbf{a}) \cdot (\mathbf{x} - \mathbf{a}) + \frac{1}{2} \left((\mathbf{x} - \mathbf{a}) H_f(\mathbf{a}) (\mathbf{x} - \mathbf{a})^T \right) \\ & + \cdots + \frac{1}{k!} \sum_{i_1, \dots, i_k=1}^k \frac{\partial^k f}{\partial x_{i_1} \dots \partial x_{i_k}}(\mathbf{a}) (x_{i_1} - a_{i_1}) \dots (x_{i_k} - a_{i_k}) \end{aligned}$$

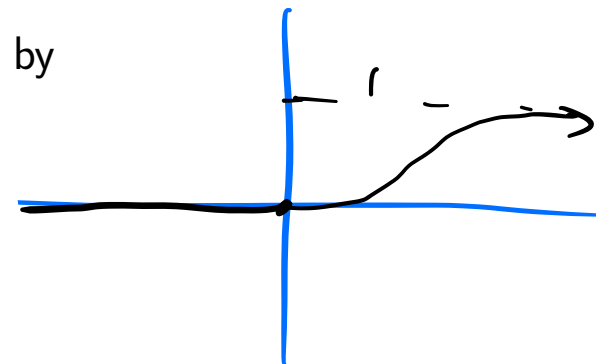
is the **Taylor polynomial of f of order k about $\mathbf{a}$.**

- If for all $\mathbf{a} \in \Omega$, there exists an open set $U_{\mathbf{a}} \ni \mathbf{a}$ such that $\lim_{k \rightarrow \infty} p_{f,k,\mathbf{a}}(\mathbf{x}) = f(\mathbf{x})$ for all $\mathbf{x} \in U_{\mathbf{a}}$, we say that f is **analytic**.
- The collection of analytic functions on Ω is denoted by $C^\omega(\Omega)$.

Smooth but not Analytic

Example. Find $f^{(n)}(0)$ for all $n \in \mathbb{Z}^+$ where $f : \mathbb{R} \rightarrow \mathbb{R}$ is defined by

$$f(x) = \begin{cases} e^{-1/x} & \text{if } x > 0, \\ 0 & \text{if } x \leq 0 \end{cases}$$



Lemma. For all $n \in \mathbb{Z}^+$, there is a polynomial P_n such that $f^{(n)}(x) = P_n\left(\frac{1}{x}\right) e^{-1/x}$ for all $x > 0$.

Proof. For $n=1$, $f'(x) = \frac{1}{x^2} e^{-1/x}$ for $x > 0$, and we can let $P_1(t) = t^2$. Now suppose that for $k \in \mathbb{Z}^+$ there is a polynomial P_k such that $f^{(k)}(x) = P_k\left(\frac{1}{x}\right) e^{-1/x}$ for all $x > 0$. Then for $x > 0$,
 $f^{(k+1)}(x) = \frac{1}{x^2} P_k\left(\frac{1}{x}\right) e^{-1/x} - \frac{1}{x^2} P_k'\left(\frac{1}{x}\right) e^{-1/x}$,
 and we can let $P_{k+1}(t) = t^2 (P_k(t) - P_k'(t))$.

Smooth but not Analytic

Eg. cont. for $n=1$, we have that $u = \frac{1}{h}$

$$\lim_{h \rightarrow 0^+} \frac{f(h) - f(0)}{h} = \lim_{h \rightarrow 0^+} \frac{e^{-1/h}}{h} = \lim_{u \rightarrow \infty} u e^{-u} = 0$$

$$\frac{e^{-1/h}}{h} = \frac{1}{h} - \frac{1}{h^2} + \frac{1}{2h^3} - \dots$$

and so $f'(0) = 0$ (left limit is trivial)

suppose that $f^{(k)}(0) \neq 0$ for some $k \in \mathbb{Z}^+$. Then

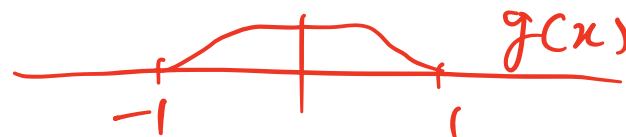
$$\lim_{h \rightarrow 0^+} \frac{f^{(k)}(h) - f^{(k)}(0)}{h} = \lim_{h \rightarrow 0^+} \frac{P_k\left(\frac{1}{h}\right) e^{-1/h}}{h}$$

for some polynomial P_k . This limit is equal to $\lim_{u \rightarrow \infty} u P_k(u) e^{-u} = 0$, so $f^{(k+1)}(0) = 0$, and the result follows by induction.

Smooth but not Analytic

Eg. cont. we see here that $f \in C^\infty(\mathbb{R})$, but the Taylor series is zero, which doesn't converge to f on any open set containing 0. So f is smooth, but not analytic.

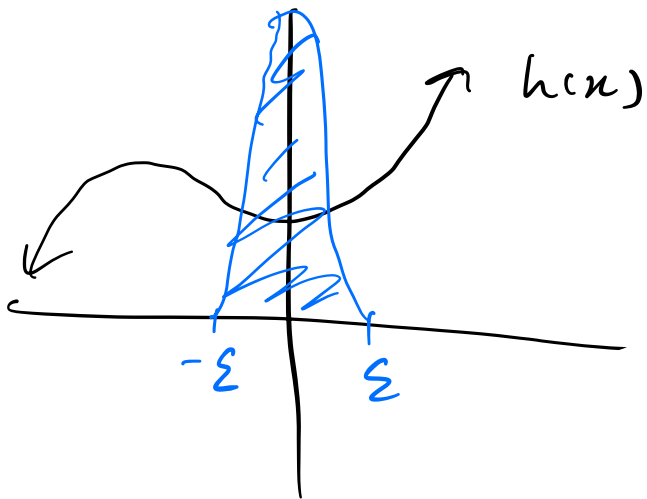
Generalise = $C^\infty(\Omega) \subsetneq C^\infty(\mathbb{R})$ for any $\emptyset \neq \Omega \subseteq \mathbb{R}^n$



$g: \mathbb{R} \rightarrow \mathbb{R}$, $g(x) = f(1-x)f(1+x)$ (can be used to make a "smooth mollifier" $I = \int_{\mathbb{R}} g(x) dx$ and $\varphi = \frac{1}{I} g$. If $h: \mathbb{R} \rightarrow \mathbb{R}$ is continuous, then

$$h(\varphi) = \lim_{\varepsilon \rightarrow 0} \frac{1}{\varepsilon} \int_{\mathbb{R}} \varphi\left(\frac{y-x}{\varepsilon}\right) h(x) dx$$

$$y = 0, \quad h(0) = \lim_{\varepsilon \rightarrow 0} \int_{\mathbb{R}} \varphi\left(\frac{x}{\varepsilon}\right) h(x) dx$$



This is how
you make sense
of Dirac Delta

Exercise

Exercise. Find the 2nd order Taylor polynomial of f about $(0,0)$ for $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by $f(x,y) = xe^{-y} + \cos(x+y)$.

$$f(0,0) = 1, \quad f_x = e^{-y} - \sin(x+y), \quad f_x(0,0) = 1$$

$$f_y = -xe^{-y} - \sin(x+y), \quad f_y(0,0) = 0$$

$$f_{xx} = -\cos(x+y), \quad f_{xx}(0,0) = -1$$

$$f_{xy} = -e^{-y} - \cos(x+y), \quad f_{xy}(0,0) = -2$$

$$f_{yy} = x^2 e^{-y} - \cos(x+y), \quad f_{yy}(0,0) = -1$$

so the 2nd order Taylor polynomial of f about $(0,0)$ is

$$1 + x - \left[\frac{1}{2} (x^2 + 4xy + y^2) \right] = (f_{xx} x^2 + 2f_{xy} xy + f_{yy} y^2)$$

Exercise

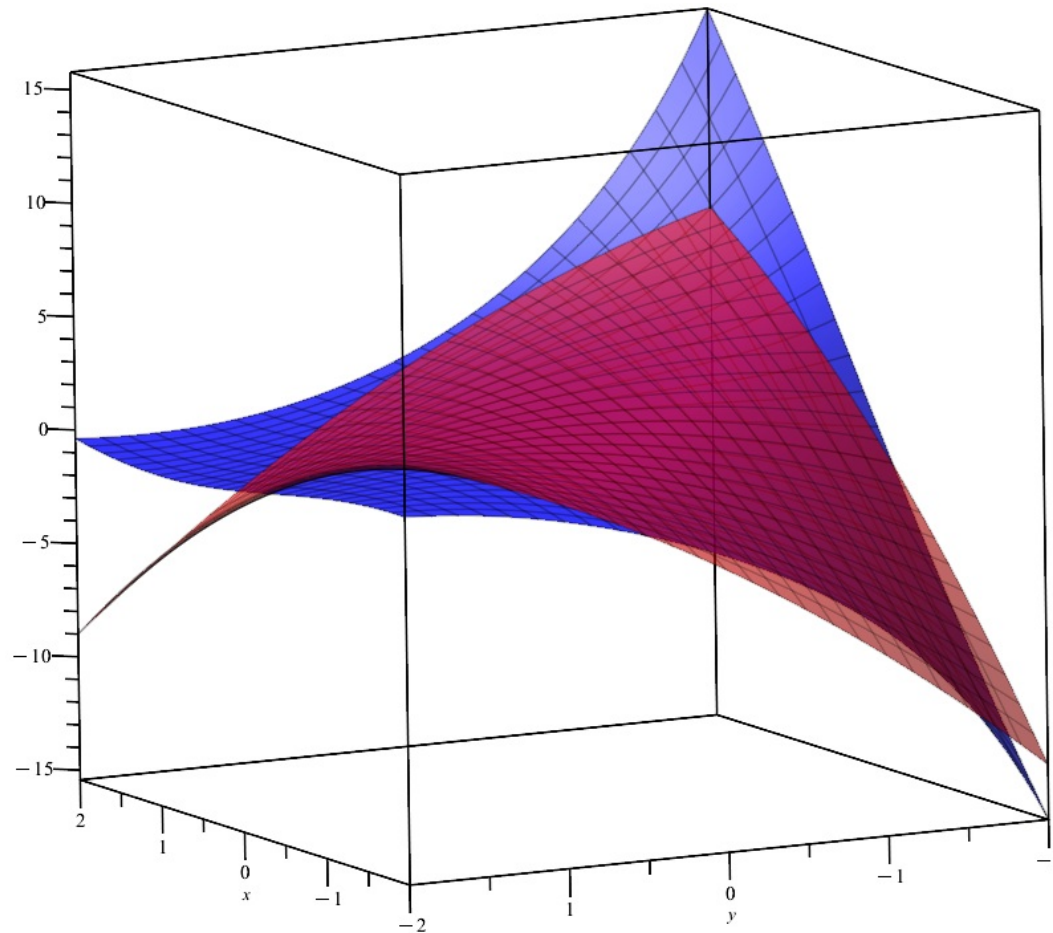


Figure: A plot of f (in blue) and its Taylor polynomial of degree 2 about $(0, 0)$ (in red).

Taylor's Theorem I

Theorem

Let $\Omega \subseteq \mathbb{R}^n$ be open and $f \in C^2(\Omega)$. Suppose that $\mathbf{x}, \mathbf{a} \in \Omega$ are such that the line segment joining $\mathbf{x}$ and $\mathbf{a}$ is contained in Ω . Then there is a point $\mathbf{z}$ on this line segment such that

$$f(\mathbf{x}) = p_{f,1,\mathbf{a}}(\mathbf{x}) + \underbrace{\frac{1}{2} ((\mathbf{x} - \mathbf{a}) H_f(\mathbf{z}) (\mathbf{x} - \mathbf{a})^T)}_{\text{remainder, like Lagrange form}}.$$

In other words,

$$f(\mathbf{x}) = f(\mathbf{a}) + \nabla f(\mathbf{a}) \cdot (\mathbf{x} - \mathbf{a}) + R_{1,\mathbf{a}}(\mathbf{x})$$

where $R_{1,\mathbf{a}}(\mathbf{x}) = \frac{1}{2} ((\mathbf{x} - \mathbf{a}) H_f(\mathbf{z}) (\mathbf{x} - \mathbf{a})^T)$ is the remainder term.

If we think of $\mathbf{a}$ as fixed and $\mathbf{x}$ varying, then $\mathbf{z}$ also depends on $\mathbf{x}$, hence why we write $R_{1,\mathbf{a}}$ as a function of $\mathbf{x}$.

By the Cauchy-Schwarz inequality, we have that

$$|R_{1,\mathbf{a}}(\mathbf{x})| \leq \frac{1}{2} |\mathbf{x} - \mathbf{a}| |H_f(\mathbf{z}) (\mathbf{x} - \mathbf{a})^T| \rightarrow 0 \text{ as } \mathbf{x} \rightarrow \mathbf{a}.$$

$$\left| \frac{R_{1,\mathbf{a}}(\mathbf{x})}{|\mathbf{x} - \mathbf{a}|} \right| \rightarrow 0 \text{ as } \underline{\mathbf{x}} \rightarrow \underline{\mathbf{a}}$$

Taylor's Theorem II

Theorem

Let $\Omega \subseteq \mathbb{R}^n$ be open and $f \in C^3(\Omega)$. Suppose that $\mathbf{x}, \mathbf{a} \in \Omega$ are such that the line segment joining $\mathbf{x}$ and $\mathbf{a}$ is contained in Ω . Then there is a point $\mathbf{z}$ on this line segment such that

$$f(\mathbf{x}) = f(\mathbf{a}) + \nabla f(\mathbf{a}) \cdot (\mathbf{x} - \mathbf{a}) + \frac{1}{2} \left((\mathbf{x} - \mathbf{a}) H_f(\mathbf{a}) (\mathbf{x} - \mathbf{a})^T \right) + R_{2,\mathbf{a}}(\mathbf{x}),$$

where $R_{2,\mathbf{a}} : \Omega \rightarrow \mathbb{R}$ is a function such that

$$\frac{|R_{2,\mathbf{a}}(\mathbf{x})|}{|\mathbf{x} - \mathbf{a}|^2} \rightarrow 0 \text{ as } \mathbf{x} \rightarrow \mathbf{a}.$$

Non-examinable:

The general version of Taylor's Theorem is almost verbatim to what is above, but we assume $f \in C^{k+1}(\Omega)$ and the conclusion is that $f(\mathbf{x}) = p_{f,k,\mathbf{a}}(\mathbf{x}) + R_{k,\mathbf{a}}(\mathbf{x})$ where

$$\frac{|R_{k,\mathbf{a}}(\mathbf{x})|}{|\mathbf{x} - \mathbf{a}|^k} \rightarrow 0 \text{ as } \mathbf{x} \rightarrow \mathbf{a}.$$

Manipulating Taylor Series

Say we want to find the Taylor polynomial for $f(x, y) = \cos(x + y^2)$ about $(0, 0)$ of degree 3. Then we could use our knowledge of one variable Taylor series:

$$\begin{aligned}\cos(x + y^2) &= 1 - \frac{(x + y^2)^2}{2} + \frac{(x + y^2)^4}{4!} - \dots \\ &= 1 - \frac{x^2}{2} - xy^2 - \frac{y^4}{2} + \dots\end{aligned}$$

so it appears that the Taylor polynomial of degree 3 should be

$$P(x, y) = 1 - \frac{x^2}{2} - xy^2$$

We can verify this since

$$f_x = -\sin(x + y^2)$$

$$f_y = -2y \sin(x + y^2)$$

$$f_{xx} = -\cos(x + y^2)$$

$$f_{yx} = f_{xy} = -2y \cos(x + y^2)$$

$$f_{yy} = -2 \sin(x + y^2) - 4y^2 \cos(x + y^2)$$

$$f_{xxx} = \sin(x + y^2)$$

$$f_{xxy} = 2y \sin(x + y^2)$$

$$f_{xyy} = 4y^2 \sin(x + y^2) - 2 \cos(x + y^2) \quad f_{yyy} = 8y^3 \sin(x + y^2) - (2 \cos(x + y^2))$$

Manipulating Taylor Series

You can check that the only non-zero values are $f_{xx}(0,0) = -1$, $f_{xyy} = -2$
So

$$\begin{aligned} P_3(x,y) &= 1 + \frac{1}{2}((-1)x^2 + 2(0)xy + (0)y^2) \\ &\quad + \frac{1}{6}((0)x^3 + 3(0)x^2y + 3(-2)xy^2 \\ &\quad + 10y^3) \\ &= 1 - \frac{x^2}{2} - xy \end{aligned}$$

as before.

Manipulating Taylor Series

The same manipulations of Taylor series you saw in first-year mathematics can be done for Taylor series in several variables:

- series for $f + g$ is the sum of the individual series;
- series for fg is the product of individual series;
- we can differentiate or integrate a series with respect to each variable.

We won't go through any proofs of these results. If you are curious, you may want to check Section 4.8.1 of the MATH1231/41 Course Pack Notes.

Example. We could find the Taylor series for $\frac{1}{1+x^2y^2}$ about $(0,0)$ by

Geometric series $\frac{1}{1+u} = 1 - u + u^2 - \dots = \sum_{k=0}^{\infty} (-1)^k u^k$ $|u| < 1$

$$\begin{aligned}\frac{1}{1+x^2y^2} &= 1 - x^2y^2 + (x^2y^2)^2 - (x^2y^2)^3 + \dots \\ &= 1 - x^2y^2 + x^4y^4 - x^6y^6 + \dots \\ &= \sum_{k=0}^{\infty} (-1)^k (xy)^{2k}\end{aligned}$$

Exercise in Manipulation

Exercise. Find the Taylor polynomial of degree two about $(1, 1)$ for $f(x, y) = \sqrt{xy^3}$.

In the vicinity of $(1, 1)$, $f(x, y) = y\sqrt{x}y$, and
The Taylor series for $x \mapsto \sqrt{x}$ at $a=1$ is

$$\sqrt{x} = 1 + \frac{x-1}{2} - \frac{(x-1)^2}{8} + \dots$$

Multiply:

$$y\sqrt{x}y = (1 + (y-1)) \left[1 + \frac{x-1}{2} - \frac{(x-1)^2}{8} \dots \right]$$

Expand

$$\left[1 + \frac{y-1}{2} - \frac{(y-1)^2}{8} \dots \right]$$

$$= \left[1 + \frac{x-1}{2} - \frac{(x-1)^2}{8} + \dots \right] \left[1 + \frac{y-1}{2} - \frac{(y-1)^2}{8} + \dots \right] \\ + \left[1 + \frac{x-1}{2} - \frac{(x-1)^2}{8} + \dots \right] \left[y-1 + \frac{(y-1)^2}{2} - \frac{(y-1)^3}{8} + \dots \right]$$

Exercise in Manipulation

Ex. cont.

$$= 1 + \frac{y-1}{2} - \frac{(y-1)^2}{8} + \frac{x-1}{2} + \frac{(x-1)(y-1)}{4} \\ + (y-1) + \frac{(y-1)^2}{2} + \frac{(x-1)(y-1)}{2} + \dots$$

$$= 1 + \frac{x-1}{2} + \frac{3(y-1)}{2} - \frac{(x-1)^2}{8} + \frac{3(x-1)(y-1)}{4} \\ + \frac{3}{8}(y-1)^2 + \dots$$

Confirm on Maple